

Solution to Tutorial 6 problems

(1)

Coordinates for P_1 : $(x, 0)$

Coordinates for P_2 : $(x + a \sin \theta, a \cos \theta)$

$$T = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \left[(\dot{x} + a \dot{\theta} \cos \theta)^2 + (-a \dot{\theta} \sin \theta)^2 \right]$$
$$= \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m \left[\dot{x}^2 + a^2 \dot{\theta}^2 + 2 \dot{x} (a \dot{\theta}) \cos \theta \right]$$

$$V = -mg a \cos \theta$$

$$L = \frac{1}{2} m (2 \dot{x}^2 + a^2 \dot{\theta}^2 + 2 a \dot{x} \dot{\theta} \cos \theta) + mg a \cos \theta$$

conjugate momenta p_x and p_θ are then given by:

$$p_x = \frac{\partial L}{\partial \dot{x}} = m (2 \dot{x} + a \dot{\theta} \cos \theta) \rightarrow (1)$$

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = m a (a \dot{\theta} + \dot{x} \cos \theta) \rightarrow (2)$$

From (1) and (2) we can obtain:

$$\dot{x} = \frac{p_x - \cos\theta (p_\theta/a)}{m(2 - \cos^2\theta)}$$

$$a\dot{\theta} = \frac{2(p_\theta/a) - \cos\theta p_x}{m(2 - \cos^2\theta)}$$

The system is conservative (Lagrangian does not have explicit time dependence)
So, the Hamiltonian is given by:

$$H(x, \theta, p_x, p_\theta) = T + V$$
$$= \frac{p_x^2 + 2(p_\theta/a)^2 - 2\cos\theta p_x (p_\theta/a)}{2m(2 - \cos^2\theta)} - mga\cos\theta$$

(after doing some tedious algebra!)

Now, $\frac{\partial H}{\partial p_x} = \frac{p_x - \cos\theta (p_\theta/a)}{m(2 - \cos^2\theta)} = \dot{x}$

$$\frac{\partial H}{\partial p_\theta} = \frac{2(p_\theta/a) - \cos\theta p_x}{m(2 - \cos^2\theta)} = \dot{\theta}$$

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(2)

In the given problem:

$$f = H(q_1, q_2, \dots, q_n; p_1, p_2, \dots, p_n; t) \\ - (\dot{q}_1 p_1 + \dot{q}_2 p_2 + \dots + \dot{q}_n p_n)$$

Now, the Euler-Lagrange eqⁿ corresponding to variable q_i is:

$$\frac{d}{dt} \left(\frac{\partial f}{\partial \dot{q}_i} \right) - \frac{\partial f}{\partial q_i} = 0$$

$$\Rightarrow \frac{d}{dt} (-p_i) - \frac{\partial H}{\partial q_i} = 0$$

$$\Rightarrow \boxed{\dot{p}_i = - \frac{\partial H}{\partial q_i}} \rightarrow (1) \quad (1 \leq i \leq n)$$

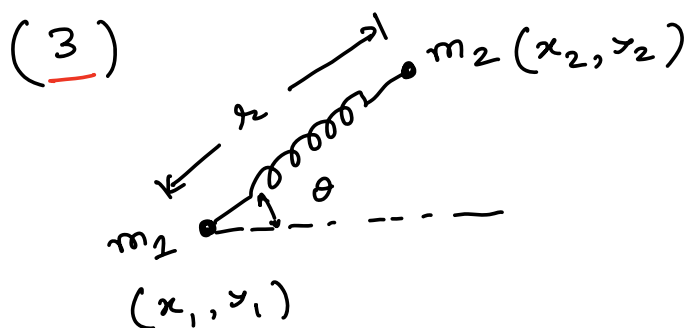
Again, the Euler-Lagrange's equation for the variable p_i is:

$$\frac{d}{dt} \left(\frac{\partial f}{\partial \dot{p}_i} \right) - \frac{\partial f}{\partial p_i} = 0$$

$$\Rightarrow \frac{d}{dt} (0) - \left(\frac{\partial H}{\partial p_i} - \dot{q}_i \right) = 0 \Rightarrow \boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}} \rightarrow (2)$$

(1) and (2) are exactly Hamilton's equations for a system with Hamiltonian $H(\vec{q}, \vec{p}, t)$.

(NOTE: There are a total of $2n$ Euler-Lagrange's equations; one for each variable!)



$$x_2 = x_1 + r \cos \theta, \quad y_2 = y_1 + r \sin \theta$$

$$T = \frac{1}{2} m_1 (\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2} m_2 (\dot{x}_2^2 + \dot{y}_2^2)$$

$$V = \frac{1}{2} k (r - l)^2$$

$$\begin{aligned} \Rightarrow (a) \quad L &= \frac{1}{2} (m_1 + m_2) (\dot{x}_1^2 + \dot{y}_1^2) + \frac{1}{2} m_2 (\dot{x}^2 + r^2 \dot{\theta}^2) \\ &\quad + m_2 r (\dot{x}_1 \cos \theta + \dot{y}_1 \sin \theta) \\ &\quad + m_2 r \dot{\theta} (\dot{y}_1 \cos \theta - \dot{x}_1 \sin \theta) - \frac{1}{2} k (r - l)^2 \end{aligned}$$

$\rightarrow (1)$

Lagrange's equations of motion are:

$$\frac{\partial L}{\partial x_1} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}_1} \right) = 0 \quad \rightarrow (2)$$

$$\frac{\partial L}{\partial y_1} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{y}_1} \right) = 0 \quad \rightarrow (3)$$

$$\frac{\partial L}{\partial z} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{z}} \right) = 0 \quad \rightarrow (4)$$

$$\frac{\partial L}{\partial \theta} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}} \right) = 0 \quad \rightarrow (5)$$

Using (1),
Eqⁿ (2) - (5) result in the followings:

- $$\frac{d}{dt} \left[(m_1 + m_2) \dot{x}_1 + m_2 \dot{z} \cos \theta - m_2 z \dot{\theta} \sin \theta \right] = 0$$

$$\Rightarrow \frac{d}{dt} (m_1 \dot{x}_1 + m_2 \dot{x}_2) = 0$$

$$\Rightarrow \frac{d}{dt} p_x = 0 \Rightarrow \boxed{p_x \equiv \text{constant}}$$

- $$\frac{d}{dt} \left[(m_1 + m_2) \dot{y}_1 + m_2 \dot{z} \sin \theta + m_2 z \dot{\theta} \cos \theta \right] = 0$$

$$\Rightarrow \frac{d}{dt} (m_1 \dot{y}_1 + m_2 \dot{y}_2) = 0$$

$$\Rightarrow \frac{d}{dt} p_y = 0 \Rightarrow \boxed{p_y \equiv \text{constant}}$$

$$\bullet \quad \frac{d}{dt} \left[m_2 \dot{r} + m_2 (\dot{x}_1 \cos \theta + \dot{y}_1 \sin \theta) \right] \\ = m_2 r \dot{\theta}^2 - k(r-l) + m_2 \dot{\theta} (\dot{y}_1 \cos \theta - \dot{x}_1 \sin \theta)$$

$$\Rightarrow \boxed{\ddot{r} - r \dot{\theta}^2 + \ddot{x}_1 \cos \theta + \ddot{y}_1 \sin \theta + \frac{k}{m_2} (r-l) = 0}$$

$$\bullet \quad \frac{d}{dt} \left[m_2 r^2 \dot{\theta} + m_2 r (\dot{y}_1 \cos \theta - \dot{x}_1 \sin \theta) \right] \\ = -m_2 \dot{r} (\dot{x}_1 \sin \theta - \dot{y}_1 \cos \theta) + m_2 r \dot{\theta} (-\dot{x}_1 \cos \theta - \dot{y}_1 \sin \theta)$$

$$\Rightarrow \boxed{\ddot{\theta} + \frac{2}{r} \dot{r} \dot{\theta} + \frac{\cos \theta}{r} \ddot{y}_1 - \frac{\sin \theta}{r} \ddot{x}_1 = 0} //$$

(b)

$$\frac{\partial L}{\partial \dot{x}_1} = p_x = \text{constant}$$

$$\frac{\partial L}{\partial \dot{y}_1} = p_y = \text{constant}$$

(c) Using L from part (a)

$$p_{x_1} = \frac{\partial L}{\partial \dot{x}_1} = (m_1 + m_2) \dot{x}_1 + m_2 \dot{r} \cos \theta - m_2 r \dot{\theta} \sin \theta$$

$$p_{y_1} = \frac{\partial L}{\partial \dot{y}_1} = (m_1 + m_2) \dot{y}_1 + m_2 \dot{r} \sin \theta - m_2 r \dot{\theta} \cos \theta$$

$$p_r = \frac{\partial L}{\partial \dot{r}} = m_2 \dot{x}_1 \cos \theta + m_2 \dot{y}_1 \sin \theta + m_2 \dot{r}$$

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = -m_2 r \dot{x}_1 \sin \theta + m_2 r \dot{y}_1 \cos \theta + m_2 r^2 \dot{\theta}$$

Doing a bit of algebra :

$$\left. \begin{aligned} \dot{x}_1 &= \frac{1}{m_1} \left[p_{x_1} - p_z \cos \theta + \frac{\sin \theta}{r} p_\theta \right] \\ \dot{y}_1 &= \frac{1}{m_1} \left[p_{y_1} - p_z \sin \theta - \frac{\cos \theta}{r} p_\theta \right] \\ \dot{z} &= \frac{1}{m_2} \left[-p_{x_1} \cos \theta - p_{y_1} \sin \theta + \frac{m_1 + m_2}{m_2} p_z \right] \\ \dot{\theta} &= \frac{1}{m_1 r} \left[p_{x_1} \sin \theta - p_{y_1} \cos \theta + \frac{m_1 + m_2}{m_2 r} p_\theta \right] \end{aligned} \right\} (6)$$

It can be seen that coordinate transformations are time independent and v is velocity independent. So,

$$H = T + V$$

Substituting the above equations [(6)] for $\dot{q}_i$ in terms of p_i we obtain:

$$\begin{aligned} H = \frac{1}{2m_1} & \left[p_{x_1}^2 + p_{y_1}^2 + \frac{m_1 + m_2}{m_2} \left(p_z^2 + \frac{p_\theta^2}{r} \right) \right. \\ & - 2p_z (p_{x_1} \cos \theta + p_{y_1} \sin \theta) \\ & + \frac{2p_\theta}{r} (p_{x_1} \sin \theta - p_{y_1} \cos \theta) \left. \right] \\ & + \frac{1}{2} K (r - b)^2 \end{aligned}$$

Hamilton's eqⁿs of motion are:

$$\dot{x}_1 = \frac{\partial H}{\partial p_{x_1}} = \frac{1}{m_1} \left[p_{x_1} - p_z \cos \theta + \frac{\sin \theta}{r} p_\theta \right]$$

$$\dot{y}_1 = \frac{\partial H}{\partial p_{y_1}} = \frac{1}{m_1} \left[p_{y_1} - p_z \sin \theta - \frac{\cos \theta}{r} p_\theta \right]$$

$$\dot{z} = \frac{\partial H}{\partial p_z} = \frac{1}{m_1} \left[\frac{m_1 + m_2}{m_2} p_z - p_{x_1} \cos \theta - p_{y_1} \sin \theta \right]$$

$$\dot{\theta} = \frac{\partial H}{\partial p_\theta} = \frac{1}{m_1 r} \left[\frac{m_1 + m_2}{m_2 r} p_\theta + p_{x_1} \sin \theta - p_{y_1} \cos \theta \right]$$

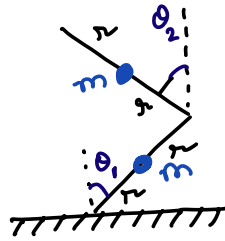
$$\dot{p}_{x_1} = - \frac{\partial H}{\partial x_1} = 0, \quad \dot{p}_{y_1} = - \frac{\partial H}{\partial y_1} = 0$$

$$\dot{p}_z = - \frac{\partial H}{\partial z} = \frac{(m_1 + m_2) p_\theta^2}{m_1 m_2 r^3} + \frac{p_\theta}{m_1 r^2} (p_{x_1} \sin \theta - p_{y_1} \cos \theta) - \kappa (r - \ell)$$

$$\begin{aligned} \dot{p}_\theta = - \frac{\partial H}{\partial \theta} &= \frac{p_z}{m_1} (-p_{x_1} \sin \theta + p_{y_1} \cos \theta) \\ &\quad - \frac{p_\theta}{m_1 r} (p_{x_1} \cos \theta + p_{y_1} \sin \theta) \end{aligned}$$

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(4)



Take θ_1 and θ_2 as the generalized coordinates, as defined in Fig. above.

The position of the bottom mass in cartesian coordinate is $(r \sin \theta_1, r \cos \theta_1)$

The position of the top mass in cartesian coordinate is
 $(2r \sin \theta_1 - r \sin \theta_2, 2r \cos \theta_1 + r \cos \theta_2)$

Thus, the potential energy of the system is:

$$\boxed{V(\theta_1, \theta_2) = mgr(3 \cos \theta_1 + \cos \theta_2)} \rightarrow (1)$$

The kinetic energy of the bottom mass is:

$$T_1 = \frac{1}{2} m r^2 \dot{\theta}_1^2$$

The K. E. of the top mass is

$$T_2 = \frac{1}{2} m r^2 \left[(2\dot{\theta}_1 \cos \theta_1 - \dot{\theta}_2 \cos \theta_2)^2 + (-2\dot{\theta}_1 \sin \theta_1 - \dot{\theta}_2 \sin \theta_2)^2 \right]$$

Now,

$$\begin{aligned} & (2\dot{\theta}_1 \cos\theta_1 - \dot{\theta}_2 \cos\theta_2)^2 \\ &= 4\dot{\theta}_1^2 \cos^2\theta_1 + \dot{\theta}_2^2 \cos^2\theta_2 - 4\dot{\theta}_1\dot{\theta}_2 \cos\theta_1 \cos\theta_2 \end{aligned}$$

Under small angle approximation

$$\cos\theta \approx 1 - \frac{\theta^2}{2}, \quad \sin\theta \approx \theta$$

So,

$$\begin{aligned} & (2\dot{\theta}_1 \cos\theta_1 - \dot{\theta}_2 \cos\theta_2)^2 \\ & \approx 4\dot{\theta}_1^2 \left(1 - \frac{\theta_1^2}{2}\right)^2 + \dot{\theta}_2^2 \left(1 - \frac{\theta_2^2}{2}\right)^2 \\ & \quad - 4\dot{\theta}_1\dot{\theta}_2 \left(1 - \frac{\theta_1^2}{2}\right) \left(1 - \frac{\theta_2^2}{2}\right) \\ &= 4\dot{\theta}_1^2 + \dot{\theta}_2^2 - 4\dot{\theta}_1\dot{\theta}_2 \end{aligned}$$

Again:

$$\begin{aligned} & (-2\dot{\theta}_1 \sin\theta_1 - \dot{\theta}_2 \sin\theta_2)^2 \\ &= 4\dot{\theta}_1^2 \sin^2\theta_1 + \dot{\theta}_2^2 \sin^2\theta_2 + 4\dot{\theta}_1\dot{\theta}_2 \sin\theta_1 \sin\theta_2 \\ &\approx 4\dot{\theta}_1^2 \theta_1^2 + \dot{\theta}_2^2 \theta_2^2 + 4\dot{\theta}_1\dot{\theta}_2 (\theta_1 \theta_2) \end{aligned}$$

Neglecting all θ_i^2 as they are too small, we can write, under small angle approximation:

$$T_2 = \frac{1}{2} m r^2 [4\dot{\theta}_1^2 + \dot{\theta}_2^2 - 4\dot{\theta}_1\dot{\theta}_2]$$

$$= \frac{1}{2} m r^2 [(2\dot{\theta}_1 - \dot{\theta}_2)^2]$$

Hence,

$$T = T_1 + T_2$$

$$= \frac{1}{2} m r^2 \dot{\theta}_1^2 + \frac{1}{2} m r^2 (2\dot{\theta}_1 - \dot{\theta}_2)^2$$

Again, under small angle approximation the potential energy is

$$V \approx mgr [3(1 - \frac{\theta_1^2}{2}) + (1 - \frac{\theta_2^2}{2})]$$

$$= mgr (4 - \frac{3}{2}\theta_1^2 - \frac{1}{2}\theta_2^2)$$

Thus, under small angle approximation

$$L(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2) \approx \frac{1}{2} m r^2 (5\dot{\theta}_1^2 - 4\dot{\theta}_1\dot{\theta}_2 + \dot{\theta}_2^2)$$

$$- mgr (4 - \frac{3}{2}\theta_1^2 - \frac{1}{2}\theta_2^2)$$

Lagrange's eqⁿs of motion:

$$\frac{\partial L}{\partial \theta_1} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_1} \right) = 0 \Rightarrow$$

$$5\ddot{\theta}_1 - 2\ddot{\theta}_2 = \frac{3g}{r} \theta_1$$

$$\frac{\partial L}{\partial \theta_2} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\theta}_2} \right) = 0 \Rightarrow$$

$$-2\ddot{\theta}_1 + \ddot{\theta}_2 = \frac{g}{r} \theta_2$$

$$5\ddot{\theta}_1 - 2\ddot{\theta}_2 = \frac{3g}{r} \theta_1$$

$$-2\ddot{\theta}_1 + \ddot{\theta}_2 = \frac{g}{r} \theta_2$$

At the instant the sticks are released we have $\theta_1 = 0$ and $\theta_2 = \epsilon$. Solving for $\ddot{\theta}_1$ and $\ddot{\theta}_2$ yields:

$$\ddot{\theta}_1 = \frac{2g\epsilon}{r} ; \quad \ddot{\theta}_2 = \frac{5g\epsilon}{r}$$

These are the required angular accelerations

$$L(\theta_1, \theta_2, \dot{\theta}_1, \dot{\theta}_2) = \frac{1}{2} m r^2 (5\dot{\theta}_1^2 - 4\dot{\theta}_1 \dot{\theta}_2 + \dot{\theta}_2^2) - m g r (4 - \frac{3}{2} \theta_1^2 - \frac{1}{2} \theta_2^2)$$

$$p_{\theta_1} = \frac{\partial L}{\partial \dot{\theta}_1} = 5 m r^2 \dot{\theta}_1 - 2 m r^2 \dot{\theta}_2$$

$$p_{\theta_2} = \frac{\partial L}{\partial \dot{\theta}_2} = -2 m r^2 \dot{\theta}_1 + m r^2 \dot{\theta}_2$$

$$\Downarrow$$

$$\dot{\theta}_1 = \frac{p_{\theta_1} + p_{\theta_2}}{m r^2} , \quad \dot{\theta}_2 = \frac{2(p_{\theta_1} + p_{\theta_2})}{m r^2}$$

$$H = \dot{\theta}_1 p_{\theta_1} + \dot{\theta}_2 p_{\theta_2} - L$$

Using straightforward algebra it can be obtained (please *verify* yourself!) :

$$H = \frac{(p_{\theta_1} + p_{\theta_2})(p_{\theta_1} + 2p_{\theta_2})}{m r^2} - \frac{(p_{\theta_1} + p_{\theta_2})^2}{2m r^2} + m g r \left(4 - \frac{3}{2} \theta_1^2 - \frac{1}{2} \theta_2^2 \right)$$

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Yes H is conserved !
 [As there is no explicit time dependence;
 $\frac{\partial H}{\partial t} = 0 \Rightarrow \frac{dH}{dt} = 0$]